caches use a special hashing function, *consistent hashing*[34] (see figure 7.14), to decide which node should store an entry.

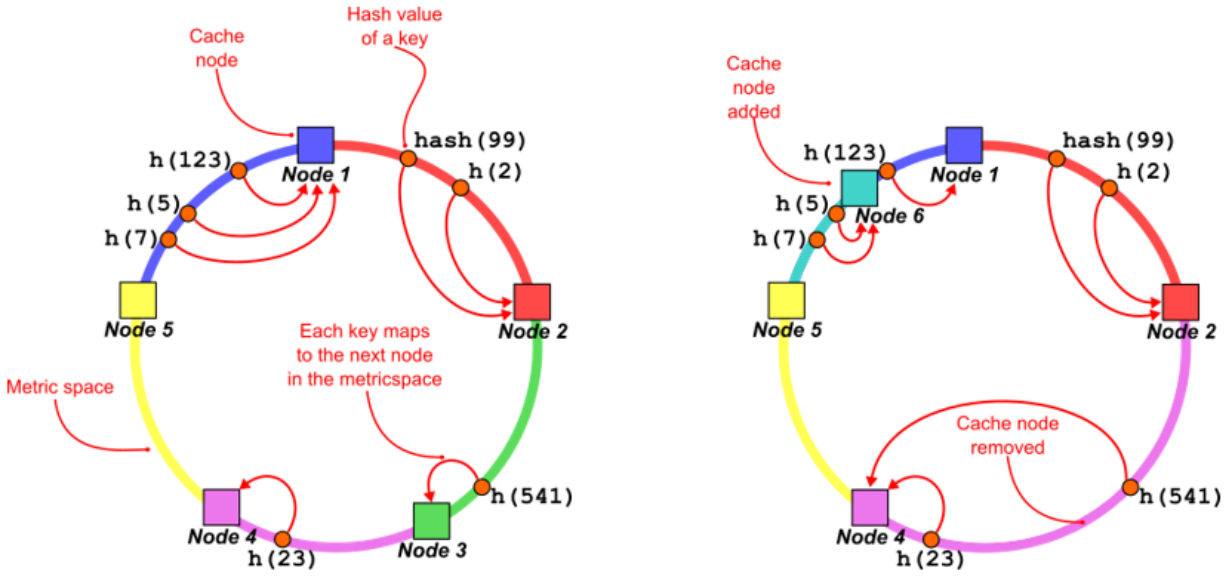


Figure 7.14 An array of cache nodes using consistent hashing for distributing keys. On the left, the initial nodes array. Node and keys map to the same metric space (the IDs space), usually represented as a circle (which could be obtained even just with a modulo operation on the hash values). Keys are assigned to the next node in the circle, i.e. the node whose value is the smallest node ID larger than the key hashing. On the right, we can see how, when a cache node is removed or added, only a fraction of the keys need to be remapped.

Web Clients also make extensive use of caching: browsers have their own caches (although those work slightly differently, storing content on your disk), but they also have a DNS cache to store the IP addresses corresponding to the domains you browse. Lately more caching has been added